

Assessing the Global Climate in 2024

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Earth had its warmest year on record while upper-ocean heat content experienced a record high.

NOAA ranks 2024 as the warmest year in its global temperature record, which dates back to 1850.

Upper ocean heat content—the amount of heat stored in the top 2000 meters of the ocean—was record high in 2024.

Antarctic sea ice extent was second lowest on record in both February (when the annual minimum occurs) and September (when the annual maximum occurs).

There were 85 named tropical storms across the globe in 2024, which was slightly below average, and 18 in the North Atlantic, which was above average.

In 2024, global surface temperature was 2.32°F (1.29°C) above the 20th-century average. This ranks as the highest global temperature in the period of 1850–2024, beating the next warmest year (2023) by 0.18°F (0.10°C). The 10 warmest years since 1850 have all occurred in the past decade. In 2024, global temperature exceeded the pre-industrial (1850–1900) average by 2.63°F (1.46°C).

Temperatures were warmer than average over the vast majority of the Earth's land surface in 2024. The largest warm anomalies were in the Arctic, northeastern North America and eastern Europe. North America, South America, Europe, Africa and Oceania had their warmest year on record. Asia and the Arctic ranked second. Overall, it was the warmest year on record for the land surface as a whole. Much of the ocean surface was warmer than average as well.

The largest warm temperature anomalies were in the North Atlantic and the western north Pacific. Overall, it was the warmest year on record for the ocean surface. Temperatures were cooler than average over some areas, such as southern Greenland, eastern Antarctica, the southeastern Pacific Ocean and the Drake Passage.

Upper ocean heat content—the amount of heat stored in the top 2000 meters of the ocean—was record high in 2024. Ocean heat content is a key climate indicator because the oceans store 90% of the excess heat in the Earth system. The indicator has been tracked globally since 1958, and there has been a steady upward trend since about 1970. The five highest values have all occurred in the last five years.

Northern Hemisphere snow cover extent averaged 9.2 million square miles in 2024, which was slightly below average. Monthly extent ranged from 18.1 million square miles in January to just 1.0 million square miles in August, both of which were slightly below average. Northern Hemisphere snow cover extent records begin in 1967.

Arctic sea ice extent averaged 4.03 million square miles in 2024, seventh lowest on record. The maximum extent in March was 5.74 million square miles, which ranked 15th lowest, while the minimum extent in September was 1.69 million square miles, which ranked sixth lowest. Global sea ice extent records began in 1979.

Antarctic sea ice extent averaged 4.00 million square miles in 2024, second lowest on record. The maximum extent in September was 6.59 million square miles, which ranked second lowest, and the

minimum extent in February was 830,000 square miles, which also ranked second lowest.

Eighty-five named storms occurred across the globe in 2024, which was slightly below the 1991–2020 average of 88. Forty-two of those reached tropical cyclone strength (≥ 74 mph), and 23 reached major tropical cyclone strength (≥ 111 mph). These also included four storms that reached Category 5 (≥ 157 mph) on the Saffir-Simpson hurricane wind scale. The global accumulated cyclone energy (ACE) was about 21% below the 1991–2020 average.

The North Atlantic had 18 named storms, which was above the 1991–2020 average of 14. Eleven of those were hurricanes, including five major hurricanes. The ACE was about 32% above normal. Most of the damage in the U.S. was attributable to two major hurricanes: Helene in September and Milton in October.

This monthly summary, developed by scientists at NOAA's National Centers for Environmental Information, is part of the suite of climate services NOAA provides to government, business, academia and the public to support informed decision-making. See our full report for a more complete summary of 2024.